THE NORMALIZATION OF A PROPER OR PROJECTIVE SCHEME IS AGAIN PROPER OR PROJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and consists of the two chains of cited stability results; no step is referred to the reader.

Editorial note. The counted spans carry 6 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-normalize-projective is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- The normalization morphism of a locally algebraic scheme is finite. Same chapter, Lemma lemma-normalization-locally-algebraic, invoked at proof lines 4959-4960. Supplies finiteness of nu.
- A finite morphism is proper and properness is stable under composition. Morphisms, Lemmas morphisms-lemma-finite-proper and morphisms-lemma-composition-proper, invoked at proof lines 4960-4962. Supplies properness of the normalization.
- A finite morphism is projective and projectivity is stable under composition. Morphisms, Lemmas morphisms-lemma-finite-projective and morphisms-lemma-composition-projective, invoked at proof lines 4963-4966. Supplies projectivity of the normalization in the projective case.

Source reference. Original lemma 0BXR: sources/stacks-varieties/pinned-original-source.tex, source lines 4903–4903. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 035N: sources/stacks-chow/tags.txt, source lines 4134–4134. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0C4P: sources/stacks-chow/tags.txt, source lines 15234–15234. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 01W3: sources/stacks-chow/tags.txt, source lines 2468–2468. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0B3I: sources/stacks-chow/tags.txt, source lines 13966–13966. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01WN: sources/stacks-chow/tags.txt, source lines 2488–2488. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k. Let $\nu : X^\nu \to X$ be the normalization morphism, see Morphisms, Definition 035N. Then X^ν is proper over k. If X is projective over k, then X^ν is projective over k.*

Proof. By Lemma 0BXR the morphism ν is finite. Hence X^ν is proper over k by Morphisms, Lemmas 01WN and 01W3. The morphism ν is projective by Morphisms, Lemma 0B3I and hence if X is projective over k, then X^ν is projective over k by Morphisms, Lemma 0C4P. □